

ARJUN CHANDRA

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INTRO

Hello! I am a Research Fellow in Electrical & Computer Engineering at Boston University, passionate about bridging human learning and AI. My work explores how insights from cognitive development can inspire new data-efficient and human-centric AI systems. I am also interested in applications of AI in neuroimaging and healthcare, with a focus on developing reliable automatic evaluation methods.

EDUCATION

Boston University

Boston, MA

B.A. Physics & Computer Science

2021 - 2025

- GPA: 3.98/4.00
- Advisors: Boqing Gong, Irving Bigio
- Relevant Coursework:
 - Multimodal Machine Learning (Graduate)
Project: Targeted Caption Generation Improves Compositional Reasoning in VLMs
 - Bandit Algorithms (Graduate)
Project: Fair Algorithms for Multi-Agent Multi-Armed Bandits
 - Quantum Computing (Graduate)
Project: Lattice-Based Key Encapsulation Mechanisms in Post-Quantum Cryptography

WORK EXPERIENCE

Adobe | San Jose, CA

May 2024 – Aug 2024

AI/ML Intern - Machine Learning Engineer

- Implemented and deployed a scalable LLM-based recommendation engine on roadmap for future release in Adobe Experience Platform AI Assistant
- Built and managed Intern Connect, a Slack app that pairs hundreds of Adobe interns around the world for weekly virtual coffee chats

RESEARCH EXPERIENCE

Boston University Data Science & Machine Learning

June 2025 – Present

Research Fellow | Advisor: Venkatesh Saligrama

- Curating synthetic data and developing automatic evaluation methods to probe failure modes in speech-to-speech models
- First-author paper submitted to EACL 2026

Boston University Image & Video Computing

Jan 2025 – Present

Research Assistant | Advisor: Boqing Gong

- Investigating developmentally plausible learning mechanisms for intelligent and data-efficient pre-training of multimodal foundation models
- Published second-author paper in ICCV 2025

Adobe Research

Aug 2024 – May 2025

Student Researcher | Advisor: Yunyao Li

- Integrated multi-source knowledge retrieval into text-to-video pipelines to achieve improved factual accuracy and interpretability
- Published first-author paper in AAAI 2025 and demonstration proposal in CVPR 2025

Boston University Biomedical Optics Lab

Aug 2023 – May 2025

Research Assistant | Advisor: Irving Bigio

- Implemented 3D object detection methods to identify myelin degeneration in microscopic brain imaging of late-stage Alzheimer's and CTE patients
- Published one paper in Alzheimer's & Dementia and one paper in Neurophotonics

PUBLICATIONS	1. Arjun Chandra, Kevin Miller, Venkatesh Ravichandran, Constantinos Papayiannis, Venkatesh Saligrama. <i>Dimension-First Evaluation of Voice Assistants: Human Chain-of-Thought and Structured Judges</i>. Submitted to (EACL) 2026. 2. Daniel Lee*, Arjun Chandra*, Yang Zhou, Yunyao Li, Simone Conia. <i>Grounding Pixels in Facts: Distilled Knowledge Retrieval for Factual Text-to-Video Generation</i>. (CVPR) 2025 Demonstration Program. 3. Shengao Wang, Arjun Chandra, Aoming Liu, Boqing Gong, Venkatesh Saligrama. <i>BabyVLM: Data-Efficient Pretraining of VLMs Inspired by Infant Learning</i>. (ICCV) 2025. 4. Daniel Lee*, Arjun Chandra*, Yang Zhou, Yunyao Li, Simone Conia. <i>Rewind and Render: Towards Factually Accurate Text-to-Video Generation with Distilled Knowledge Retrieval</i>. (AAAI) 2025 Demonstration Program. 5. Anna Novoseltseva, Gulce Kureli, Shuaibin Chang, Jiarui Yang, Precious D. Antinew, Arjun Chandra, Samer Berghol, Elizabeth Spurlock, Ann C. McKee, Bertrand R. Huber, Hui Wang, David A. Boas, Irving J. Bigio. <i>Imaging myelin degradation in ex vivo pre-frontal cortex tissue blocks in Alzheimer’s disease and chronic traumatic encephalopathy</i>. (Alzheimer’s & Dementia). 6. Anna Novoseltseva, Arjun Chandra, Alexander J. Gray, Shuying Li, Mikayla Bradsby, Irving J. Bigio. <i>Accelerating Myelin Defect Detection in Neurodegenerative Disorders: A Human-in-the-Loop Deep Learning Approach with Birefringence Microscopy</i>. (Neurophotonics).	
TEACHING EXPERIENCE	• Teaching Assistant, CS 460 (Database Systems) • Course Assistant, CS 460 (Database Systems) • Learning Assistant, PY212 (General Physics 2) • Learning Assistant, PY252 (Electricity & Magnetism) • Peer Tutor, Educational Resource Center	Spring 2025 Fall 2024 Fall 2023 Spring 2023 2022 - 2024
AWARDS & HONORS	• Faculty Award for Interdisciplinary Majors, Boston University • UROP Student Research Award, Boston University • SAEF Award, Boston University • Third Place, MIT iQuHack	May 2025 Jan 2024 - May 2025 Dec 2024 Feb 2024
LEADERSHIP & OUTREACH	Boston University PRISM Peer Mentor • Mentored incoming class of physics majors in course planning, research, and other campus opportunities Boston University Children’s Center Classroom Assistant • Assisted with daily routines and educational activities for preschool children Boston University Music Engagement Treasurer • Coordinated with Allocations Board to manage semester budget and submit budget requests	Aug 2024 – May 2025 Jan 2024 - Aug 2024 Sept 2023 – May 2024